Calculus 3 Discussion Assignments

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Date

Contents

Discussion 1 2
- 12.1 2
- 12.2 4
- 12.3 7
- 12.4 9

Discussion 2 15

Discussion 1

12.1

Question 12:

Find the lengths of the sides of the triangle PQR. Determine whether the triangle is right, isosceles, or neither.

$$P = (2, -1, 0) \quad Q = (4, 1, 1) \quad R(4, -5, 4)$$

Solution:

$$\begin{gathered}
\ell : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R} \\
(x_1, x_2, x_3), (y_1, y_2, y_3) \mapsto \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2} \\
\ell(P, Q) = \sqrt{(2-4)^2 + (-1-1)^2 + (0-1)^2} \\
= \sqrt{4 + 4 + 1} = \sqrt{9} = 3 \\
\ell(Q, R) = \sqrt{(4-4)^2 + (1+5)^2 + (1-4)^2} \\
\sqrt{0 + 36 + 9} = \sqrt{45} \\
\ell(P, R) = \sqrt{(2-4)^2 + (-1+5)^2 + (0-4)^2} \\
= \sqrt{4 + 16 + 16} = \sqrt{36} = 6 \\
6^2 + 3^2 = 45 \\
\sqrt{45}^2 = 45
\end{gathered}$$

Therefore this is a right triangle.

Question 16:

Find the equation of the sphere with center $(2, -6, 4)$ and radius 5. Describe its intersection with each of the coordinate planes.

Solution:

The equation of a sphere with center (h, k, l) and radius r is given as

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

Thus, we have

$$(x - 2)^2 + (y + 6)^2 + (z - 4)^2 = 25$$

For all points intersecting a coordinate plane, either x, y, or z must be equal to 0.

$$\text{Suppose } x = 0$$

$$\Rightarrow 4 + (y+6)^2 + (z-4)^2 = 25$$
$$\Rightarrow (y+6)^2 + (z-4)^2 = 21$$

The intersection of the sphere with the yz-plane is described by the circle $(y+6)^2 + (z-4)^2 = 21$.

$$\text{Suppose } y = 0$$
$$\Rightarrow (x-2)^2 + 36 + (z-4)^2 = 25$$
$$\Rightarrow (x-2)^2 + (z-4)^2 = -11$$

Notice that $\forall r \in \mathbb{R}(r^2 \geq 0) \therefore \forall(x, z \in \mathbb{R})\left((x-2)^2 + (z-4)^2 \geq 0\right)$. Thus, this equation only has complex-valued solutions. It's implied that $x, y, z \in \mathbb{R}$, thus the sphere does not intersect the xz-plane unless complex solutions are allowed.

$$\text{Suppose } z = 0$$
$$\Rightarrow (x-2)^2 + (y+6)^2 + 16 = 25$$
$$\Rightarrow (x-2)^2 + (y+6)^2 = 9$$

The intersection of the sphere with the xy-plane is described by the circle $\Rightarrow (x-2)^2 + (y+6)^2 = 9$.

Question 18:

Find the equation of the sphere that passes through the origin and whose center is $(1, 2, 3)$.

Solution:

The equation of a sphere with center (h, k, l) and radius r is given as

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

Thus, we have

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = r^2$$

All points on the sphere are the same distance from the center, thus the radius must equal the distance between the origin and the center of the sphere.

$$r = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1+4+9} = \sqrt{14}$$

Therefore the sphere must be defined as

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = 14$$

Question 46:

Write inequalities to describe the solid upper hemisphere of the sphere with radius 2 and centered at the origin.

Solution:

This can be done with a single inequality. Generally the square root operator $\sqrt{a}$ with $a \in \mathbb{R}^+$ defines the *positive* square root, so by taking the square root of both sides of the equation of a circle we obtain the upper half of the sphere. Thus, we obtain

$$r \leq \sqrt{(x-h)^2 + (y-k)^2 + (z-l)^2}$$

We use $\leq$ here to obtain all radii less than or equal to the maximum radius of this upper half of the sphere, which will define the entire area. Plugging in the center $(h, k, l) = (0, 0, 0)$ and $r = 2$, we find

$$2 \leq \sqrt{x^2 + y^2 + z^2}$$

12.2

Question 21:

Find $\vec{a} + \vec{b}$, $3\vec{a} + 2\vec{b}$, $\|\vec{a}\|$, and $\|\vec{a} - \vec{b}\|$.

$$\vec{a} = 4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}, \quad \vec{b} = 2\hat{\mathbf{i}} - 4\hat{\mathbf{k}}$$

Solution:

$$\begin{aligned}
\vec{a} + \vec{b} =& (4+2)\hat{\mathbf{i}} + (-3+0)\hat{\mathbf{j}} + (2-4)\hat{\mathbf{k}} \\
=& 6\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}} \\
4\vec{a} + 2\vec{b} =& (16+4)\hat{\mathbf{i}} + (-12+0)\hat{\mathbf{j}} + (8-8)\hat{\mathbf{k}} \\
=& 20\hat{\mathbf{i}} - 12\hat{\mathbf{j}} \\
\|\vec{a}\| =& \sqrt{4^2 + (-3)^2 + 2^2} \\
=& \sqrt{16+9+4} \\
=& \sqrt{29} \\
\|\vec{a} - \vec{b}\| =& \sqrt{(4-2)^2 + (-3-0)^2 + (2+4)^2} \\
=& \sqrt{4+9+36} \\
=& \sqrt{49} \\
=& 7
\end{aligned}$$

Question 22:

Find $\vec{a} + \vec{b}$, $3\vec{a} + 2\vec{b}$, $\|\vec{a}\|$, and $\|\vec{a} - \vec{b}\|$.

$$\vec{a} = \langle 8, 1, -4 \rangle, \quad \vec{b} = \langle 5, -2, 1 \rangle$$

Solution:

$$\begin{aligned}
\vec{a}+\vec{b} =& \langle 8+5, 1-2, -4+1\rangle \\
=& \langle 13, -1, -3\rangle \\
4\vec{a}+2\vec{b} =& \langle 32+10, 4-4, -16+2\rangle \\
=& \langle 42, 0, -14\rangle \\
\|\vec{a}\| =& \sqrt{64+1+16} \\
=& \sqrt{81} \\
=& 9 \\
\|\vec{a}-\vec{b}\| =& \sqrt{(8-5)^2+(1+2)^2+(-4-1)^2} \\
=& \sqrt{9+9+25} \\
=& \sqrt{43}
\end{aligned}$$

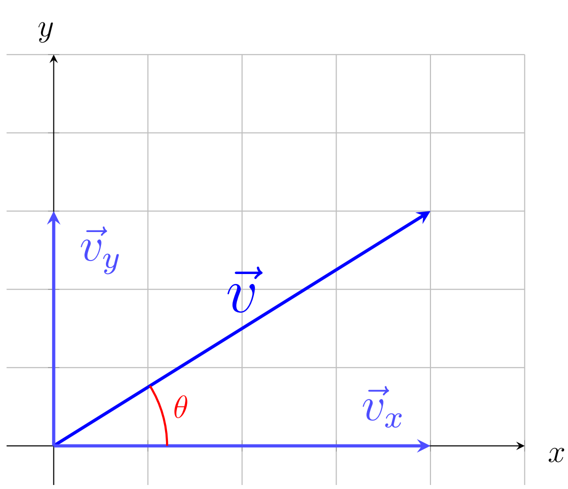
Question 28:

Find the angle formed between the given vector and the positive direction of the x-axis.

$$8\hat{\mathbf{i}}+6\hat{\mathbf{j}}$$

Solution:

Let $\vec{v} = 8\hat{\mathbf{i}}+6\hat{\mathbf{j}} = \langle 8, 6\rangle$. We already know the x and y components of the vector, and we know the vector is within the first quadrant of $\mathbb{R}^2$, as illustrated below. It follows that $\theta = \arctan\left(\frac{6}{8}\right) = 67.5° \approx 0.205\pi$



Question 30:

A child pulls a sled through the snow on a level path with a force of 50N exerted at an angle of 38° with the horizontal, find the horizontal and vertical components of the force.

Solution:

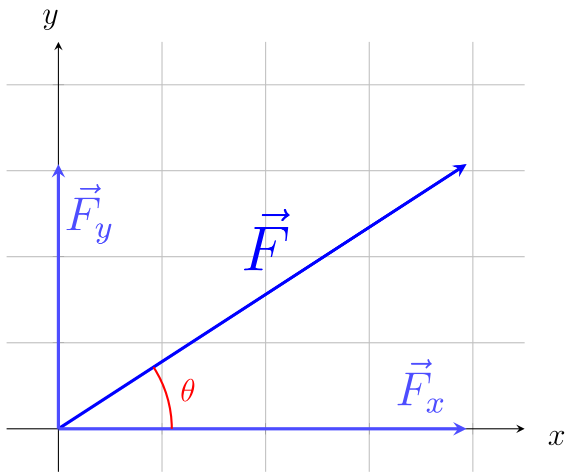
It follows from the geometric description of this problem, as seen below, that

$$\vec{F} = \|\vec{F}\|\hat{\mathbf{i}}\cos\theta + \|\vec{F}\|\hat{\mathbf{j}}\sin\theta$$
$$\Rightarrow \vec{F} = \langle \|\vec{F}\|\cos\theta, \|\vec{F}\|\sin\theta \rangle$$

We now plug in known values.

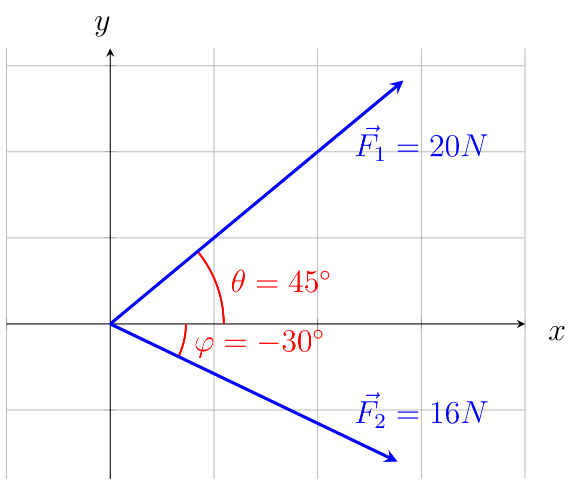
$$\Rightarrow \vec{F} = \langle 50\cos(38°), 50\sin(38°) \rangle$$

It follows that $\vec{F}_x = 50\cos(38°)\hat{\mathbf{i}}$ and $\vec{F}_y = 50\sin(38°)\hat{\mathbf{j}}$. The magnitudes of the components are approximately 39.401N and 30.783N for the x and y components, respectively.



Question 32:

Find the magnitude of the resultant force and the angle formed with the positive x-axis.



Solution:

From the geometric description of this problem, it follows that

$$\vec{F}_1 = \|\vec{F}_1\|\hat{\mathbf{i}}\cos\theta + \|\vec{F}_1\|\hat{\mathbf{j}}\sin\theta$$

$$\vec{F_2} = \|\vec{F_2}\|\hat{\mathbf{i}}\cos\varphi + \|\vec{F_1}\|\hat{\mathbf{j}}\sin\varphi$$
$$\Rightarrow \vec{F_1} + \vec{F_2} = (\|\vec{F_1}\|\cos\theta + \|\vec{F_2}\|\cos\varphi)\hat{\mathbf{i}} + (\|\vec{F_1}\|\sin\theta + \|\vec{F_2}\|\sin\varphi)\hat{\mathbf{j}}$$

We plug in $\|\vec{F_1}\| = 20$, $\|\vec{F_2}\| = 16$, $\theta = 45$, and $\varphi = -30$. Recall that $\cos(45) = \sin(45) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$, $\cos(-30) = \frac{\sqrt{3}}{2}$, and $\sin(-30) = -\frac{1}{2}$.

$$\Rightarrow \vec{F_1} + \vec{F_2} = \left(\frac{20\sqrt{2}}{2} + \frac{16\sqrt{3}}{2}\right)\hat{\mathbf{i}} + \left(\frac{20\sqrt{2}}{2} - \frac{16}{2}\right)\hat{\mathbf{j}}$$
$$= (10\sqrt{2} + 8\sqrt{3})\hat{\mathbf{i}} + (10\sqrt{2} - 8)\hat{\mathbf{j}}$$
$$\approx 27.999\hat{\mathbf{i}} + 6.142\hat{\mathbf{j}}$$

12.3

Question 20:

Find the angle formed between the vectors $\vec{a} = 8\hat{\mathbf{i}} - \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ and $\vec{b} = 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

Solution:

The dot product of two vectors $\vec{b}, \vec{a} \in V_n$ is given as

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\theta$$

where θ is the angle between the vectors. We can solve for θ.

$$\Rightarrow \cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|\|\vec{a}\|}$$
$$\Rightarrow \theta = \arccos\left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|\|\vec{a}\|}\right)$$

Recall $\vec{a} = 8\hat{\mathbf{i}} - \hat{\mathbf{j}} + 4\hat{\mathbf{k}} \wedge \vec{b} = 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$.

$$\Rightarrow \theta = \arccos\left(\frac{\langle 8, -1, 4\rangle \cdot \langle 0, 4, 2\rangle}{\sqrt{8^2 + (-1)^2 + 4^2} \cdot \sqrt{0^2 + 4^2 + 2^2}}\right)$$
$$= \arccos\left(\frac{-4 + 8}{\sqrt{81 \cdot 20}}\right)$$
$$= \arccos\left(\frac{4}{18\sqrt{5}}\right) \approx 84.3^\circ$$

The question specifies the approximation is to be taken to the nearest degree.

$$\lfloor 84.3^\circ \rfloor = 84^\circ$$

Question 24:

Determine whether the given vectors are orthogonal, parallel, or neither.

$$\vec{u} = \langle -5, 4, -2 \rangle, \quad \vec{v} = \langle 3, 4, -1 \rangle$$

Solution:

$$\begin{aligned} &\Rightarrow \vec{u} \cdot \vec{v} = -15 + 16 + 2 = 3 \\ \|\vec{u}\|\|\vec{v}\| &= \sqrt{(-5)^2 + 4^2 + (-2)^2} \cdot \sqrt{3^2 + 4^2 + (-1)^2} \\ &= \sqrt{1170} \neq 3 \end{aligned}$$

Therefore these vectors are neither orthogonal nor parallel.

$$\vec{u} = 9\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 3\hat{\mathbf{k}} \quad \vec{v} = -6\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

Solution:

$$\begin{aligned} &\Rightarrow \vec{u} \cdot \vec{v} = -54 - 24 - 6 = -84 \\ \|\vec{u}\|\|\vec{v}\| &= \sqrt{9^2 + (-6)^2 + 3^2} \cdot \sqrt{(-6)^2 + 4^2 + (-2)^2} \\ &= \sqrt{7056} = 84 \\ &\therefore \vec{u} \| \vec{v} \end{aligned}$$

Therefore these vectors are parallel.

$$\vec{u} = \langle c, c, c \rangle \quad \vec{v} = \langle c, 0, -c \rangle$$

Solution:

$$\Rightarrow \vec{u} \cdot \vec{v} = c^2 - c^2 = 0$$

Therefore these vectors are perpendicular.

Reasoning:

Two vectors $\vec{u}, \vec{v} \in V_n$ are orthogonal if an inner product $\langle \vec{u}, \vec{v} \rangle = 0$. A dot product is an example of an inner product for $\mathbb{R}^n$, so we may use the dot product to prove or disprove orthogonality. Furthermore, the dot product can be defined as $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$, where θ is the angle between the vectors. Any two parallel lines have $\theta \in \{0, \pi\}$, and $\cos 0 = 1$ and $\cos\pi = -1$. Thus, the dot product of any two parallel vectors will be equal to the products of the magnitudes of the vectors, or the negative of the magnitudes if the vectors point in opposite directions. That is, $\vec{u} \| \vec{v} \Leftrightarrow \vec{u} \cdot \vec{v} = \pm \|\vec{u}\|\|\vec{v}\|$.

Question 40:

Find the scalar and vector projections of $\vec{b}$ onto $\vec{a}$.

Solution:

The scalar projection of one vector $\vec{b} \in V_n$ onto $\vec{a} \in V_n$ is given as

$$\text{comp}_{\vec{a}}\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}$$

Let $\vec{a} = \langle 1, 4 \rangle, \quad \vec{b} = \langle 2, 3 \rangle$. The scalar projection is calculated as follows.

$$\frac{2 + 12}{\sqrt{1^2 + 4^2}} = \frac{14}{\sqrt{17}}$$

The vector projection of one vector onto another is given as

$$\begin{aligned}
\text{proj}_{\vec{a}}\vec{b} &= \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\vec{a} \\
&= \frac{14}{17}\langle 1, 4 \rangle \\
&= \left\langle \frac{14}{17}, \frac{56}{17} \right\rangle
\end{aligned}$$

Question 44:

Find the scalar and vector projections of $\vec{b}$ onto $\vec{a}$.

Solution:

The scalar projection of one vector $\vec{b} \in V_n$ onto $\vec{a} \in V_n$ is given as

$$\text{comp}_{\vec{a}}\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}$$

Let $\vec{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}, \quad \vec{b} = 5\hat{\mathbf{i}} - \hat{\mathbf{k}}$. The scalar projection is calculated as follows.

$$\frac{5 - 3}{\sqrt{1^2 + 2^2 + 3^2}} = \frac{2}{\sqrt{14}}$$

The vector projection of one vector onto another is given as

$$\begin{aligned}
\text{proj}_{\vec{a}}\vec{b} &= \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\vec{a} \\
&= \frac{2}{14}\langle 1, 2, 3 \rangle \\
&= \left\langle \frac{2}{14}, \frac{4}{14}, \frac{6}{14} \right\rangle
\end{aligned}$$

12.4

Question 13:

State whether the following expressions are meaningful. If not, explain why. If so, state whether it outputs a scalar or vector quantity.

$$\vec{a} \cdot (\vec{b} \times \vec{c})$$

Solution:

This statement outputs a scalar.

$$\vec{a} \times (\vec{b} \cdot \vec{c})$$

Solution:

This statement is meaningless as it is not defined. The dot product maps two vectors into a scalar,.thus leaving the equation as the cross product of a scalar and a vector. The cross product is a mapping between two vectors into another vector, and thus is not defined between vectors and scalars.

$$\vec{a} \times (\vec{b} \times \vec{c})$$

Solution:

This statement outputs a vector.

$$\vec{a} \cdot (\vec{b} \cdot \vec{c})$$

Solution:

This statement is meaningless as it is not defined. The dot product maps two vectors into a scalar, thus leaving the equation as the dot product of a scalar and a vector. The dot product is not defined between scalars and vectors.

$$(\vec{a} \cdot \vec{b}) \times (\vec{c} \cdot \vec{d})$$

Solution:

The dot product maps two vectors into a scalar, and thus the equation is the cross product of two scalars. The cross product is a of between two vectors into a third vector, and thus is not defined between scalars.

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$$

Solution:

This statement outputs a scalar.

Question 14:

Find $\|\vec{u} \times \vec{v}\|$ and determine whether $\vec{u} \times \vec{v}$ is directed into or out of the page. θ is the angle between the vectors.

$$\|\vec{u}\| = 10 \quad \|\vec{v}\| = 8 \quad \theta = 60^\circ$$

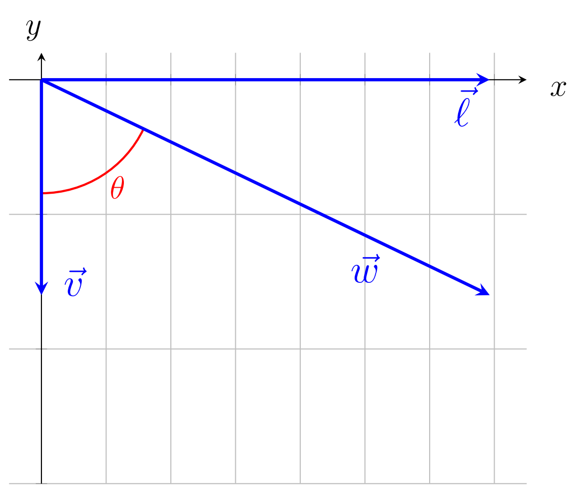
Solution:

The cross product between two vectors is defined

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} a_1 & 1_3 \\ b_1 & b_3 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} a_1 & 1_2 \\ b_1 & b_2 \end{vmatrix} \hat{\mathbf{k}}$$

From the picture, it is reasonable to assume $\vec{v} = -\|\vec{v}\|\hat{\mathbf{j}}$. If this assumption happens to be wrong, it does not in fact change the answer. We then define a new vector, as illustrated in the picture and formally defined underneath, where $\vec{\ell}$ is orthogonal to $\vec{v}$ and remains within the same plane, and is of such a magnitude that a right triangle can be formed between the vectors with the angle $\theta = 60^\circ$. We also define $\vec{w}$ as being the vector of such a magnitude as to complete the right triangle, that is $\vec{w} = \vec{\ell} + \vec{v}$.



$$\exists(\vec{\ell} \in \mathbb{R}^2) \left((\vec{\ell} \cdot \vec{v} = 0) \land \left(\tan\theta = \frac{\|\vec{\ell}\|}{\|\vec{v}\|} \right) \right)$$

$$\text{Let } \vec{w} = \langle \|\vec{\ell}\|, \|\vec{v}\| \rangle$$

From these definitions, we obtain a vector $\vec{w}$ and its components, wherein the direction of $\vec{w}$ is equal to that of $\vec{u}$, but the magnitude is unknown. We can solve for the x component of the vector and then find the unit vector $\hat{w}$. Since $\hat{w}$ points in the same direction as $\vec{u}$, it will follow that $10\hat{w} = \vec{u}$. We will notice that since we defined $\vec{\ell}$ to satisfy $\tan\theta = \frac{\|\vec{\ell}\|}{\|\vec{v}\|}$, we can solve for $\|\vec{\ell}\|$.

$$\begin{aligned} \tan\theta &= \frac{\|\vec{\ell}\|}{\|\vec{v}\|} \\ \Rightarrow \|\vec{\ell}\| &= \|\vec{v}\|\tan\theta \end{aligned}$$

Recall $\theta = 60^\circ$ and $\|\vec{v}\| = 8$.

$$\begin{aligned} \tan 60^\circ &= \sqrt{3} \\ \Rightarrow \|\vec{\ell}\| &= 8\sqrt{3} \\ \therefore \vec{w} &= 8\sqrt{3}\hat{\mathbf{i}} - 8\hat{\mathbf{j}} \end{aligned}$$

Now we solve for the unit vector $\hat{w}$.

$$\begin{aligned} \hat{w} &= \frac{\vec{w}}{\|\vec{w}\|} \\ &= \frac{8\sqrt{3}\hat{\mathbf{i}} - 8\hat{\mathbf{j}}}{\sqrt{(8\sqrt{3})^2 + (-8)^2}} \\ &= \frac{8\sqrt{3}\hat{\mathbf{i}} - 8\hat{\mathbf{j}}}{\sqrt{256}} \\ &= \frac{1}{16}(8\sqrt{3}\hat{\mathbf{i}} - 8\hat{\mathbf{j}}) \end{aligned}$$

We solve for $\vec{u}$.

$$\begin{aligned} 10\hat{w} &= \vec{u} \\ \Rightarrow \vec{u} &= \frac{10}{16}(8\sqrt{3}\hat{\mathbf{i}} - 8\hat{\mathbf{j}}) \\ &= 5\sqrt{3}\hat{\mathbf{i}} - 5\hat{\mathbf{j}} \\ &= \langle 5\sqrt{3}, -5\rangle \end{aligned}$$

We can now compute the cross product $\vec{u} \times \vec{v}$.

$$\begin{aligned} \vec{u} \times \vec{v} = &\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \\ = &\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}\hat{\mathbf{i}} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}\hat{\mathbf{j}} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}\hat{\mathbf{k}} \\ =&(u_2v_3 - v_2u_3)\hat{\mathbf{i}} - (u_1v_3 - v_1u_3)\hat{\mathbf{j}} + (u_1v_2 - v_1u_2)\hat{\mathbf{k}} \\ =&((-5\cdot 0) - (-8\cdot 0))\hat{\mathbf{i}} - ((5\sqrt{3}\cdot 0) - (0\cdot 0))\hat{\mathbf{j}} + ((-8\cdot 5\sqrt{3}) - (0\cdot(-5)))\hat{\mathbf{k}} \\ =& -40\sqrt{3}\hat{\mathbf{k}} \end{aligned}$$

We can now answer the original question.

$$\begin{aligned}\|\vec{u} \times \vec{v}\| &= \|\langle 0, 0, -40\sqrt{3}\rangle\| \\ &= 40\sqrt{3}\end{aligned}$$

We can verify this using the fact that $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\| \sin\theta$.

$$\begin{aligned}\|\vec{u}\|\|\vec{v}\| \sin(60^\circ) &= 10 \cdot 8 \cdot \frac{\sqrt{3}}{2} \\ &= 40\sqrt{3}\end{aligned}$$

Thus, our answer is correct.
Since our cross product is equal to $\langle 0, 0, -40\sqrt{3}\rangle$, and thus the z component of the vector is negative, it will be directed into the page. This can be verified trivially with the right hand rule, if the reader so desires.

Question 28:

Find the area of the parallelogram with vertices $P = (1, 0, 2)$, $Q = (3, 3, 3)$, $R = (7, 5, 8)$, $S = (5, 2, 7)$.

Solution:

From the geometric interpretation of the cross product in *link theorem*, we have

$$A = \|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin\theta$$

We can define $\overrightarrow{PQ} = \vec{a}$ and $\overrightarrow{PS} = \vec{b}$.

$$\begin{aligned}\vec{a} = \overrightarrow{PQ} &= \langle 3 - 1, 3 - 0, 3 - 2\rangle = \langle 2, 3, 1\rangle \\ \vec{b} = \overrightarrow{PS} &= \langle 5 - 1, 2 - 0, 7 - 2\rangle = \langle 4, 2, 5\rangle\end{aligned}$$

We recall from the definition of the dot product that

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \|\vec{a}\|\|\vec{b}\| \cos\theta \\ \Rightarrow \cos\theta &= \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|\|\vec{b}\|}\end{aligned}$$

We can use geometric reasoning to not have to calculate θ. Since $\cos\theta$ is defined as the adjacent leg of a right triangle divided by the hypotenuse, we find that $\vec{a} \cdot \vec{b}$ must be the adjacent leg and $\|\vec{a}\|\|\vec{b}\|$ must be the hypotenuse. $\sin\theta$ is defined as the opposide leg divided by the hypotenuse, and the opposite leg can be calculated via the pythagorean theorem.

$$\begin{aligned}a^2 + b^2 &= c^2 \\ \Rightarrow a^2 + (\vec{a} \cdot \vec{b})^2 &= (\|\vec{a}\|\|\vec{b}\|)^2 \\ \Rightarrow a &= \sqrt{(\|\vec{a}\|\|\vec{b}\|)^2 - (\vec{a} \cdot \vec{b})^2}\end{aligned}$$

It follows from the above statements that

$$\Rightarrow \sin\theta = \frac{\sqrt{(\|\vec{a}\|\|\vec{b}\|)^2 - (\vec{a}\cdot\vec{b})^2}}{\|\vec{a}\|\|\vec{b}\|}$$

Recall that the formula for area is given as

$$A = \|\vec{a}\|\|\vec{b}\|\sin\theta$$

$$\Rightarrow A = \|\vec{a}\|\|\vec{b}\|\frac{\sqrt{(\|\vec{a}\|\|\vec{b}\|)^2 - (\vec{a}\cdot\vec{b})^2}}{\|\vec{a}\|\|\vec{b}\|}$$

$$\Rightarrow A = \sqrt{(\|\vec{a}\|\|\vec{b}\|)^2 - (\vec{a}\cdot\vec{b})^2}$$

We can now plug in values and calculate.

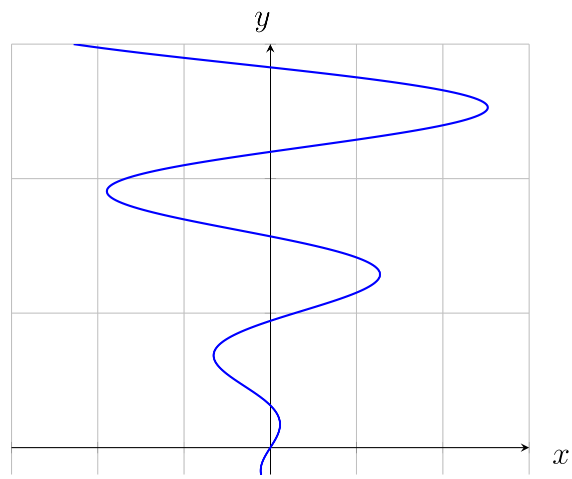
$$\begin{aligned}\Rightarrow A =& \sqrt{\left(\sqrt{2^2+3^2+1^2}\cdot\sqrt{4^2+2^2+5^2}\right)^2 - (2\cdot 4 + 3\cdot 2 + 1\cdot 5)^2}\\ =& \sqrt{(4+9+1)(16+4+25) - (8+6+5)^2}\\ =& \sqrt{(15\cdot 45) - 19^2}\\ =& \sqrt{675-361}\\ =& \sqrt{314}\end{aligned}$$

Discussion 2

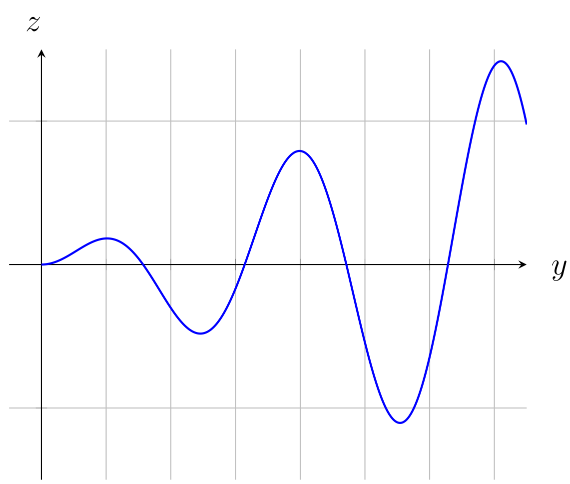
Questions 13.1.25-30:

25. $x = t\cos(t), \quad y = t, \quad z = t\sin(t), \quad t \geq 0$
Intuitively, this space curve appears to match the graph II, but there's an interesting way to verify this. We can consider the two dimensional parametric relations obtained by considering only two variables. Since we can project a single point in $\mathbb{R}^3$ onto a coordinate plane by setting one of its components equal to 0, we can think of this as projecting an entire space curve onto each plane to analyze it. We can consider the following graphs:

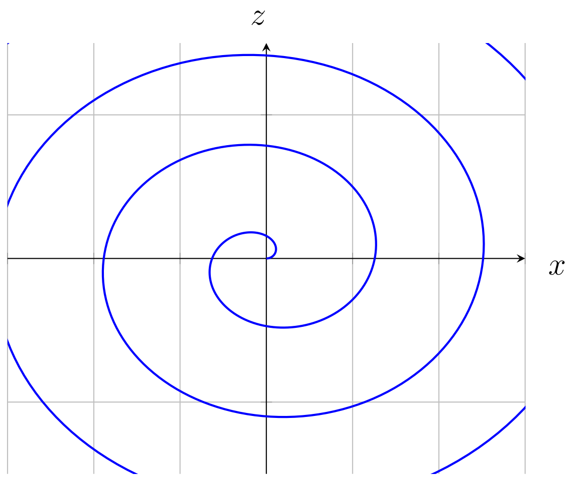
$$x = t\cos(t), \quad y = t$$



$$y = t, \quad z = t\sin(t)$$



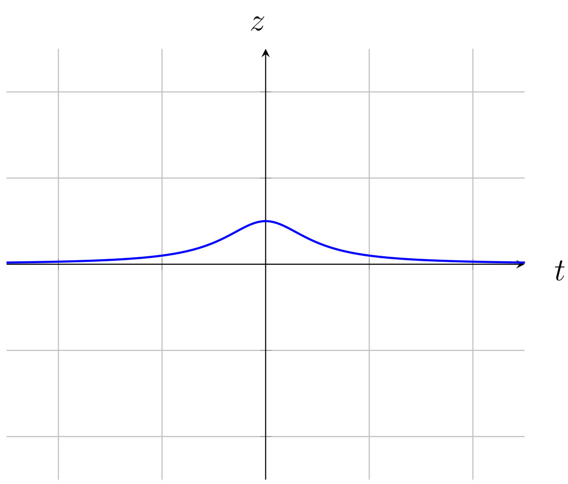
$$x = t\cos(t), \quad z = t\sin(t)$$



This is exactly what we would see if we were able to manipulate this function and look at it from one of its sides, so this confirms that the given space curve matches the graph II.

26. $x = \cos(t), \quad y = \sin(t), \quad z = \frac{1}{1+t^2}$
We can recall from both discussions in prior classes and knowledge from calculus 2 that $x = \cos(t)$, $y = \sin(t)$ is the parameterization of the unit circle. Thus, we simply want to understand how $z = \frac{1}{1+t^2}$ will effect the behavior of this graph. There are no parameters placed on t for this problem, which implies $t \in \mathbb{R}$. Since $\forall t \in \mathbb{R}(t^2 \in \mathbb{R}^+ \cup \{0\})$, this means that the behavior of z for $t \leq 0$ is identical to when $t \geq 0$, however the unit circle will spin the opposite direction for $t \leq 0$. Thus, we need to identify the behavior of z, then identify a graph matching this behavior for both the forward and reverse directions of the unit circle. Since $t \in \mathbb{R}$, we can simply consider a graph of z as a function of t to get a graphical intuition for the behavior of this function about the z-axis. I have included one for simplicity, but I'm sure we could analyze it without.

$$z = \frac{1}{1+t^2}$$



We can see clearly here that this function peaks at $z = 1$ for $t = 0$, and then quickly approaches $z = 0$, which can be trivially verified as an asymptote of the function:

$$\lim_{t \to \infty} \frac{1}{1+t^2} = 0$$

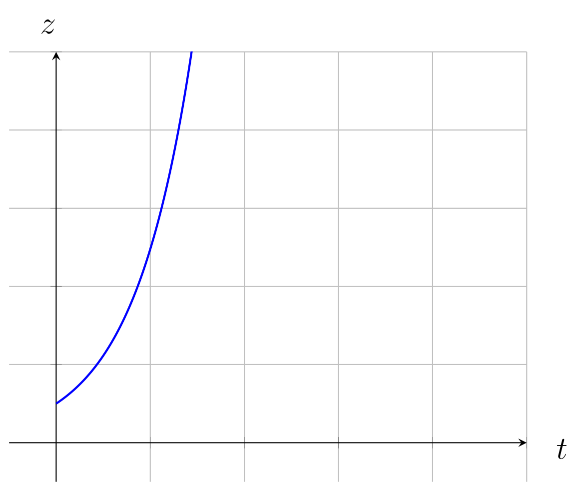
and we showed earlier that the behavior of this function for $t \leq 0$ is equivalent to that of $t \geq 0$. Graph VI matches every single criteria that has been layed out herein. It has a peak in z which rapidly approaches $z = 0$, it's of a circular form in the xy-plane, and the circle revolves in both directions due to allowing $t \in \mathbb{R}$. Thus, $x = \cos(t),\ y = \sin(t)\ z = \frac{1}{1+t^2}$ is the parameterization of the space curve depicted in graph VI.

27. $x = t, \quad y = \frac{1}{1+t^2}, \quad z = t^2$
From our analysis of the previous graph, we know how y will behave. Since t does not have any bounds specified, and thus it's implied that $t \in \mathbb{R}$, we know that x and z will go to ∞. However, we know that $\lim_{t\to-\infty} x = -\infty$, and $\lim_{t\to\infty} x = \infty$. However, $\lim_{t\to-\infty} z = \lim_{t\to\infty} z = \infty$, since $z = t^2$, and as stated earlier, $\forall t \in \mathbb{R}(t^2 \in \mathbb{R}^+ \cup \{0\})$. Graph V fits this description, as y is at a peak for $t = 0$ and approaches 0 for $t \neq 0$; x and y both go off to infinity, but only x goes to $-\infty$ as well as ∞, whereas z only goes to ∞. Thus, $x = t,\ y = \frac{1}{1+t^2},\ z = t^2$ is the parameterization of the space curve depicted in graph V.

28. $x = \cos(t), \quad y = \sin(t), \quad z = \cos(2t)$
We know that $x = \cos(t)$ and $y = \sin(t)$ is a parameterization of the unit circle, so we consider the behavior of z. There are no given bounds for t, so we consider $t \in \mathbb{R}$. We know that $\forall t \in \mathbb{R}(\cos(2t) \in [-1, 1])$, so this space curve must form a closed loop of some shape, and the only closed graph that has not been identified already is graph I. Thus, this is the parameterization of graph I.

29. $x = \cos(8t), \quad y = \sin(8t), \quad z = e^{0.8t}, \quad t \geq 0$
We have seen previously that $x = \cos(t)$ and $y = \sin(t)$ parameterize the unit circle. From precalc we also know the amplitude of a sinusoidal function of will not change by multiplying the variable inside the function by a quantity, it will only change the frequency of oscillation. Thus, this will still parameterize a unit circle in the xy-plane, but we must also consider the behavior of z. For $t = 0$, we have $z = e^0 = 1$. This establishes a lower bound for z, and as t approaches ∞, z will also approach ∞. This can be visualized with this graph:

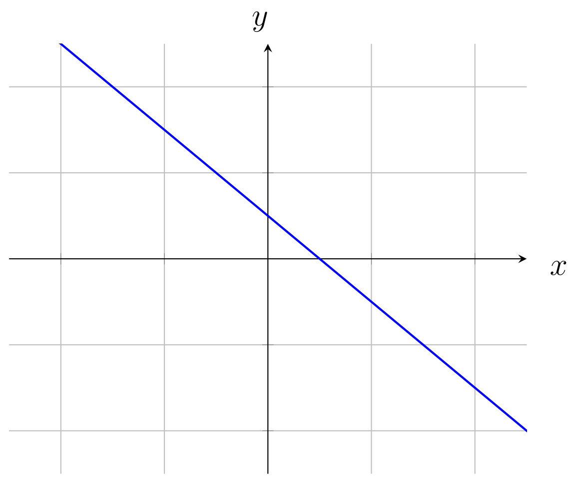
$z = e^{0.8t}$



As we can see, it's clearly approaching ∞ at an exponential rate. The graph that follows this same behavior is graph IV. The space curve therein is a unit circle that is approaching ∞ in z at an exponential rate. Thus, this is the parameterization of graph IV.

30. $x = \cos^2(t), \quad y = \sin^2(t), \quad z = t$

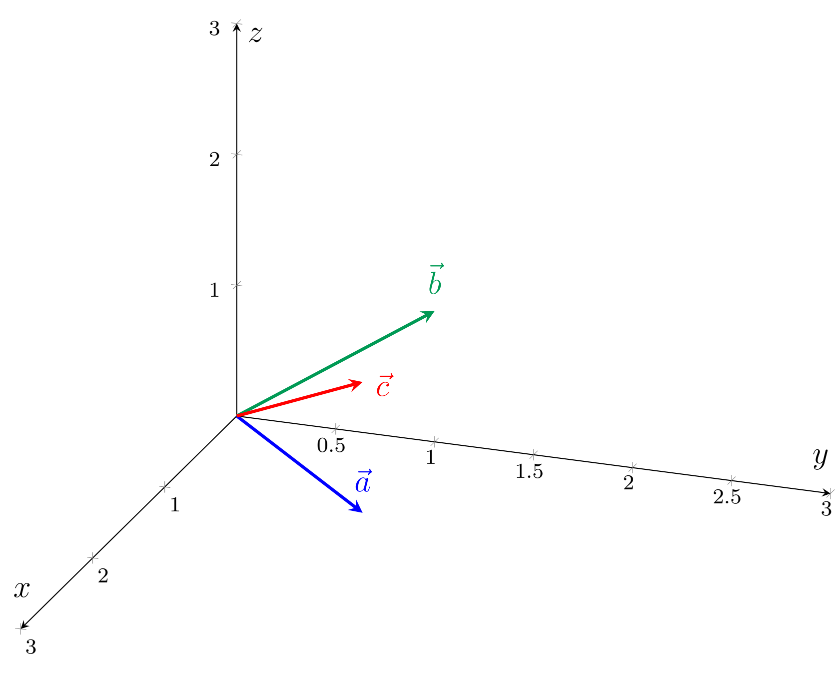
This one will clearly match with graph III, since it's the only unused graph. However, we can still prove this. Since $\forall t \in \mathbb{R}(\sin(t) \in [-1, 1])$, and the same also holds for $\cos(t)$, we will observe that $\sin^2(t), \cos^2(t) \in [0, 1]$. Thus, the graph cannot have any negative x or y values. Furthermore, we know $\sin^2 \vartheta + \cos^2 \vartheta = 1$, and we can also consider the graph of $x + y = 1 \Leftrightarrow y = 1 - x$:



This graph forms a diagonal line across the xy-plane, and since $x + y = 1$ is equivalent to saying $\cos^2(t) + \sin^2(t) = 1$, since this is a true statement, we must also find our x and y coordinates constrained within a subset of the same diagonal, where $x, y \in [0, 1]$. However, since cos and sin take on different values for different t values, we will see a wave as we progress about the z-axis. This description perfectly matches the graph III, thus proving that these equations parameterize the space curve thereof.

Question 12.4.34

Find the volume of the parallelpiped given by the vectors $\vec{a} = \hat{\mathbf{\imath}} + \hat{\mathbf{\jmath}}$, $\vec{b} = \hat{\mathbf{\jmath}} + \hat{\mathbf{k}}$, $\vec{c} = \hat{\mathbf{\imath}} + \hat{\mathbf{\jmath}} + \hat{\mathbf{k}}$



We know from class that the volume V is equivalent to the area of the base times the height. We can express this as $\left|\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)\right|$, and it doesn't matter which vectors we choose to take the cross product of, since any two vectors form a parallelogram. It also doesn't matter what order we give the cross product in, since we are looking for an absolute value, and we know $\vec{v}\times\vec{u} = -\left(\vec{u}\times\vec{v}\right)$, and $|-a| = |a|$ since $|\cdot| : \mathbb{R}\to\mathbb{R}^+\cup\{0\}$. It will then follow that

$$\begin{aligned}
V &= \left|\vec{a}\cdot(\vec{b}\times\vec{c})\right| \\
&= \left|\langle 1,1,0\rangle\cdot\left(\begin{vmatrix}1 & 1\\ 1 & 1\end{vmatrix}\hat{\imath} - \begin{vmatrix}0 & 1\\ 1 & 1\end{vmatrix}\hat{\jmath} + \begin{vmatrix}0 & 1\\ 1 & 1\end{vmatrix}\hat{k}\right)\right| \\
&= \left|\begin{vmatrix}1 & 1\\ 1 & 1\end{vmatrix}\hat{\imath} - \begin{vmatrix}0 & 1\\ 1 & 1\end{vmatrix}\hat{\jmath}\right| \\
&= |(1-1)-(0-1)| \\
&= |1| \\
&= 1
\end{aligned}$$

The volume of the parallelpiped is thus 1unit^3.